



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, UG Programmes CE,CSE,ECE,EEE,IT & ME are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Regulation: R20		II / IV - B.Tech. I - Semester							
ELECTRONICS & COMMUNICATION ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20 BS 2102	Numerical Methods and Advanced Calculus	BS	3	3	0	0	30	70	100
B20 EC 2101	Electronic Devices and Circuits	PC	3	3	0	0	30	70	100
B20 EC 2102	Digital Logic Design	PC	3	3	0	0	30	70	100
B20 EC 2103	Signals and Systems	PC	3	3	0	0	30	70	100
B20 EC 2104	Random Variables and Stochastic Processes	PC	3	3	0	0	30	70	100
B20 EC 2105	Electronic Devices and Circuits Lab With Simulation	PC	1.5	0	0	3	15	35	50
B20 EC 2106	Digital Logic Design Lab	PC	1.5	0	0	3	15	35	50
B20 EC 2107	Signals And Systems Lab	PC	1.5	0	0	3	15	35	50
B20 EC 2108	ARDIUNO programming	SOC	2	1	0	2	--	50	50
B20 MC 2101	Environmental Science	MC	0	2	0	0	--	--	--
B20 MC 2103	English proficiency	MC	0	2	0	0	--	--	--
TOTAL			21.5	20	0	11	195	505	700

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2102	BS	3	--	--	3	30	70	3 Hrs.
NUMERICAL METHODS AND ADVANCED CALCULUS								
(Common to ECE & ME)								
Pre-requisites: Basic concepts of calculus.								
Course Objectives: Students are expected to learn								
1	Numerical methods to solve algebraic and transcendental equations. The concept of interpolation and its use for equally and unequally spaced data points.							
2	Methods for numerical evaluation of integrals and for solving first order ODEs.							
3	Concepts of double, triple integrals and its applications.							
4	Concepts of Gradient, divergence, curl and vector integral theorems.							
5	Concept of Analytic function, its implications and simple applications in Electrostatics and fluid flow problems.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Find a real root of algebraic and transcendental equations. Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.							K3
2.	Evaluate numerically certain definite integrals and solve a first order ordinary differential equation by Euler and RK methods.							K3
3.	Evaluate double integrals and determine the areas.							K3
4.	Evaluate triple integrals and determine the volumes.							K3
5.	Find the gradient of a scalar function, divergence and curl of a vector function. Solve simple problems using vector integral theorems.							K3
6.	Comprehend the concept of Analytic function and apply in Electrostatics and Fluid dynamics							K3
SYLLABUS								
UNIT-I (10Hrs)	Solution of Algebraic and Transcendental Equations: Introduction, Bisection method, Method of false position, Iteration method & Newton-Raphson method. Interpolation: Introduction, forward differences, backward differences and Central differences. Differences of a polynomial, Newton's forward and backward interpolation formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange interpolation formulae.							
UNIT-II (14 Hrs)	Numerical Integration and solution of Ordinary Differential equations: Trapezoidal rule, Simpson's 1/3 rd rule. Solution of first order ordinary differential equations by Taylor's method, Picard's method, Euler's method, Modified Euler's method, Fourth order Runge-Kutta method.							

UNIT-III (10 Hrs)	Multiple integrals: Evaluation of Double integrals, change of order of integration, Change of variables and areas. Evaluation of Triple integrals and volumes.
UNIT-IV (12 Hrs)	Vector calculus: Scalar and vector point functions, Vector Differentiation, Gradient, Directional derivative, Divergence, Curl, Scalar Potential. Line integral, Work done, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof).
UNIT-V (12 Hrs)	Functions of a Complex Variable: Definition, limit and continuity of a function of a complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$. The Bilinear transformation.
Text Books:	
1.	Scope and Treatment as in “Higher Engineering Mathematics”, by Dr.B.S.Grewal, 43 rd Edition, Khanna Publishers.
Reference Books:	
1.	Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley
2.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
3.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
4.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.
5.	Higher Engineering Mathematics, by Dr. M.K.Venkatraman, the NationalPublishingCompany.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2101	PC	3	--	--	3	30	70	3 Hrs
ELECTRONIC DEVICES AND CIRCUITS								
(Common to ECE & EEE)								
Course Objectives:								
1.	To analyze the operating principles, characteristics and electrical parameters of Diode, BJT, JFET and MOSFET.							
2.	To illustrate the concepts of biasing in BJT, JFET and MOSFET.							
3.	To analyze amplifier circuits using equivalent circuits.							
4.	To illustrate the application of diodes in rectifiers and regulated power supply.							
5.	To analyze the frequency response of small signal amplifiers.							
Course Outcomes:At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Analyze the characteristics and operation of Diode, BJT.							K4
2.	Deduce the stability factors of different biasing circuits of BJT.							K3
3.	Analyze the characteristics and operation of JFET and MOSFET.							K4
4.	Design the small signal BJT single stage amplifiers.							K4
SYLLABUS								
UNIT-I (12 Hrs)								
Semiconductor diode and its applications: Potential variation in graded semiconductors, Mass action law, Open circuited PN junction, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, characteristics of Tunnel diode and Zener diode. Half wave, Full wave and Bridge Rectifiers with and without filters (inductor, capacitor, L-section and π -section).								
UNIT-II (8 Hrs)								
Bipolar junction transistors: Operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α, β, γ relation between α, β and γ early effect or base-width modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.								
UNIT-III (8 Hrs)								
Transistor Biasing Circuits: Need for biasing, The operating point, reasons for shifting of operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , V_{BE} , & β . Bias compensation, thermal runaway, thermal stability.								

UNIT-IV (8 Hrs)	Field Effect transistors: JFET construction and its characteristics, pinch off voltage, FET as a VVR, FET small signal model, MOSFET and its characteristics.
UNIT-V (12 Hrs)	Transistors at low and High frequencies: Transistor hybrid model, H-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.
Text Books:	
1.	Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.
2.	Electronic Devices and Circuits: N Salivahanan and Suresh Kumar, Third edition, TMH.
Reference Books:	
1.	Electronic Devices and Circuits Theory, Boylsted, 10th Edition, Pearson.
2.	Electronic Principles: Albert Paul Malvino, McGraw-Hill.
e-Resources:	
1.	https://nptel.ac.in/courses/117/103/117103063/
2.	https://www.iare.ac.in/sites/default/files/lecture_notes/IARE_ECE_EDC%20NOTES.pdf

Code	Category	L	T	P	C	I.M	E.M	Exam	
B20EC2102	PC	3	--	--	3	30	70	3Hrs	
DIGITAL LOGIC DESIGN									
(For ECE)									
Course Objectives:									
1.	To provide insights of various number systems, Boolean functions and logic gates.								
2.	To introduce different minimization techniques.								
3.	To explain the design process of various combinational and sequential circuits.								
4.	To introduce the design of synchronous and asynchronous sequential circuits.								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								KL
1.	Convert one number system to another, analyze logic gates and Boolean theorems.								K3
2.	Analyze digital circuits using different minimization techniques.								K3
3.	Design various combinational and sequential circuits along with applications.								K4
4.	Design counters and state machines by applying the knowledge of synchronous and asynchronous sequential circuits.								K4
SYLLABUS									
UNIT-I (10 Hrs)	Number Systems, Codes and Boolean Algebra: Number Systems, Base Conversion Methods, Complements of Numbers, 4 Bit Codes-BCD, Excess-3, 2421, 8421 codes. Even and Odd parity, Hamming code, Error detecting and Error correcting codes. Fundamentals of Boolean Algebra and Logic Gates, Boolean theorems and proofs.								
UNIT-II (8 Hrs)	Boolean Functions and Minimization: Boolean SOP and POS functions-Canonical and Standard. Realization with Universal Gates– Simplification of Boolean functions using Karnaugh Map (up to 6 variables) and Quine McClusky methods.								
UNIT-III (8 Hrs)	Combinational Logic Circuits and Design: Logic Design of Combinational circuits – Binary Adder, Subtractor, Decoders, Encoders, Multiplexers, Demultiplexers, Code Convertors, Priority Encoders, Seven – segment Displays, 4-bit digital Comparators and realization of Boolean functions withPLDs.								
UNIT-IV (10 Hrs)	Sequential Logic Circuits and Design: RS & SR latches, clocked RS, SR, JK, T & D Flip-flops with pre-set and clear inputs. Race around problem, MS-JK-FF, Excitation tables of all Flip- Flops and conversions from one type to another. Design of Shift Registers with SIPO, SISO, PIPO and PISO modes and universal shift register. Design of Ring counters and Johnson counters.								
UNIT-V (10 Hrs)	Asynchronous and Synchronous Sequential Circuits: Design of Asynchronous counters (Ripple counters) for any modulus. Design of Synchronous counters using SR, JK, T and D-								

	Flip Flops. Analysis and Design of Synchronous circuits with State Diagrams and State Reduction. Design of sequence detectors and generators. Basics of Asynchronous Sequential Circuits.
Text Books:	
1.	Digital Design, Morris Mano, PHI, 3rd Edition,2001
2.	Switching and Finite Automata Theory, 2nd Edition, ZviKohavi, Tata McGraw-Hill, 1978
Reference Books:	
1.	Fundamentals of Digital Circuits by A.Anand Kumar, PHI learning Pvt. Ltd.
2.	Fundamentals of Logic Design, charles H Roth, Thomson Publications, 5th edition, 2009
e-Resources:	
1.	https://www.geeksforgeeks.org/minimization-of-boolean-functions/
2.	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/#blg
3.	https://www.cs.ou.edu/~fagg/classes/ame3623_s05/lectures/class_sequential.pdf

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2103	PC	3	--	--	3	30	70	3Hrs
SIGNALS AND SYSTEMS								
(For ECE)								
Course Objectives:								
1.	To introduce the fundamental concepts and techniques associated with the understanding of signals and systems.							
2.	To familiarize with techniques suitable for analysing both continuous-time and discrete time LTI systems using transforms.							
3.	To familiarize with development of the mathematical skills to solve problems involving convolution, filtering, and sampling.							
Course Outcomes:At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Outline the basic concepts of signals and systems.							K2
2.	Analyze the spectral characteristics of Continuous Time and Discrete Time periodic and aperiodic signals using Fourier analysis.							K4
3.	Analyze system properties based on impulse response and Fourier analysis.							K4
4.	Apply Laplace- transforms for analysing Continuous -time signals and systems.							K3
5.	Apply Z- transforms for analysing discrete-time signals and systems.							K3
6.	Outline the process of sampling and the effects of under sampling.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Continuous –Time and Discrete –Time signals and systems Continuous –Time and Discrete –Time signals, Signal Energy and Power, Periodic Signals, Even and odd Signals, continuous- Time complex Exponential and Sinusoidal Signals, Discrete –Time complex Exponential and Sinusoidal Signals and their Periodicity , The Unit Impulse and Unit step Functions, The Continuous-Time Unit impulse and Unit step Sequence, Continuous –Time and Discrete –Time Systems, Interconnections of Systems, Basic System Properties, Continuous–Time and Discrete Time LTI Systems: – The Graphical interpretation of Convolution Integral and The Convolution Sum, Causal LTI Systems Described by Differential and Difference Equations, Singularity Functions.							
UNIT-II (8 Hrs)	Fourier Series Representation of Periodic Signals Introduction, Fourier Series Representation of continuous time Periodic Signals (Complex Exponential and Trigonometric Fourier Series only), convergence of the Fourier Series, Properties of continuous time Fourier Series, Fourier Series representation of discrete time periodic signals, Properties of discrete time Fourier Series (Elementary Level on DTFS).							

UNIT-III (8 Hrs)	Continuous and Discrete time Fourier Transform Introduction, Representation of Aperiodic signals, The continuous time Fourier Transform, The Fourier Transform for periodic signals, Properties of the continuous time Fourier Transform, Systems characterized by linear constant coefficient differential equations. Discrete time Fourier Transform, Properties of the Discrete time Fourier Transform, Systems characterized by linear constant coefficient differential equations (Elementary Level on DTFT).
UNIT-IV (6 Hrs)	Laplace Transform Introduction, The Laplace Transform, the region of convergence for Laplace Transforms, The Inverse Laplace Transform, Properties of Laplace Transforms, The initial and Final value theorems, Analysis and characterization of LTI systems using the Laplace Transforms.
UNIT-V (8 Hrs)	Sampling Theorem and Z-transform Introduction to Sampling Theorem, Statement of Sampling Theorem for Low pass and Band pass signals (Theorem Proof for Low Pass signals only), reconstruction of a signal from its samples using interpolation, discussion on Oversampling, Critical sampling and Under sampling (aliasing). The Z-Transform (Bilateral and unilateral), The Inverse Z-Transform, Properties of Z-Transform, Initial and Final Value theorems, some common Z-transform pairs, Analysis and characterization of LTI systems using the Z-Transforms.
Text Books:	
1.	Signals Systems and Communication-B. P. Lathi, BS Publication.
2.	Signals and Systems- Alan V. Oppenheim, Alan S. Willsky and Ian T. Young, PHI, 2ndEdn.
Reference Books:	
1.	Signals and Systems – P.RamakrishnaRao, TMH.
2.	Signals and Systems- A.AnandaKumar,PHI.
e-Resources:	
1.	https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/video-lectures/
2.	https://swayam.gov.in/nd1_noc20_ee06/preview

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2104	PC	3	--	--	3	30	70	3 Hrs.
RANDOM VARIABLES AND STOCHASTIC PROCESSES								
(For ECE)								
Course Objectives:								
1.	To introduce the fundamental concepts and theorems of probability theory.							
2.	To introduce fundamental concepts of random variables and their statistical descriptions							
3.	To discuss various types of random processes, their properties, spectral representation and applications.							
4.	To be aware of the difference between time averages and statistical averages, Stationary and Ergodicity.							
5.	To apply elements of stochastic processes for solving problems in real life and identify the concept of Noise as applicable to linear Systems							
Course Outcomes:At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Demonstrate the axiomatic formulation of modern probability theory.							K2
2.	Characterize Probability Models and functions of Random variables based on single and multiple random variables.							K3
3.	Evaluate and apply moments and characteristic functions and acquire the concept of Inequalities and probabilistic limits.							K4
4.	Assimilate the concept of Random process and determine covariance and spectral density of stationary random processes.							K2
5.	Identify specific applications to Poisson and Gaussian processes, and Analyze the response of random inputs to linear time invariant systems.							K4
SYLLABUS								
UNIT-I (7 Hrs)		Probability Theory: Definitions of Probability, Axioms of Probability, Probability Spaces, Properties of Probabilities, Joint and Conditional Probabilities, Independent Events						
UNIT-II (11 Hrs)		Random Variables: Probability Distribution Function, Probability Density Function, Joint Distribution of Two Variables, Conditional Probability Distribution and Density, Independent Random Variables, Normal Distribution, Cauchy’s distribution, Exponential Distribution, Binomial Distribution, Poisson distribution, Functions of Random Variables.						
UNIT-III (9 Hrs)		Statistical Averages: Random Vectors, Statistical Averages, Characteristic Function of Random Variables, Inequalities of Chebyshev” s and Schwartz, Convergence Concepts, Central Limit Theorem.						
UNIT-IV (9 Hrs)		Random Processes :Introduction, Definitions, Stationarity, Ergodicity, Covariance Function and their Properties, Spectral Representation, Wiener- Khinchin Theorem						

UNIT-V (9 Hrs)	Linear Systems and Random Noise Processes : Classification of Linear systems, Response of Linear Systems to Random signals, Spectral characteristics of system Response, Gaussian processes, Poisson Processes, Low-pass and Bandpass Noise Representation.
Text Books:	
1.	Probability Theory and Random Signal Principles, Peebles, Tata McGraw Hill Publishers.
2.	Probability Theory and Random Processes, S. P. Eugene Xavier, S. Chand and Co. New Delhi, 1998 (2 nd Edition).
Reference Books:	
1.	B.P. Lathi, “Signals, Systems & Communications”, B.S. Publications, 2003.
2.	Signal Analysis, Papoulis, McGraw Hill N. Y., 1977.
e-Resources:	
1.	https://nptel.ac.in/courses/111/102/111102111/
2.	https://nptel.ac.in/courses/117/105/117105085/

Code	Category	L	T	P	C	I.M	E.M	Exam	
B20EC2105	PC	--	--	3	1.5	15	35	3 Hrs	
ELECTRONIC DEVICES AND CIRCUITS LAB WITH SIMULATION									
(Common to ECE & EEE)									
Course Objectives:									
1.	To analyse the modelling, characteristics and electrical parameters of Diode, BJT, JFET and MOSFET.								
2.	To illustrate the concepts of biasing in BJT, JFET and MOSFET.								
3.	To illustrate the application of diode in rectifiers and regulated power supply.								
4.	To analyze single stage amplifier circuits using equivalent circuits.								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								KL
1.	Apply the concepts of different electronic devices to verify their characteristics and measure the important parameters.								K3
2.	Analyze the performance of rectifier circuits with and without filters.								K4
3.	Analyze the performance of BJT and FET amplifier circuits.								K4
4.	Simulation and Design of small electronic circuits using BJT and FET.								K4
HARDWARE									
1	Study and Analyze V-I Characteristics of Semiconductor Diode (Ge & Si), LED and Zener Diode.								
2	Determination of Ripple Factor and Regulation Characteristics of Half Wave and Full Wave Rectifier With and Without Filter.								
3	Study and Analyze the Characteristics of BJT in CE Configuration and determination of h-parameters.								
4	Study and Analyze the Characteristics of BJT in CB Configuration and determination of h-parameters.								
5	Study and Analyze the JFET Characteristics.								
6	Design of Biasing Circuits for BJT and FET.								
7	Design of Common Emitter amplifier using BJT.								
8	Frequency response of Common Source FET amplifier.								
SOFTWARE									
1.	Simulation of V-I Characteristics of Semiconductor Diode, LED and Zener Diode.								
2.	Simulation of Regulation Characteristics of Zener Diode.								
3.	Simulation of CC Amplifier.								
4.	Simulation of JFET Characteristics.								

5.	Simulation of BJT Characteristics in CE Configuration.
6.	Simulation of BJT Characteristics in CB Configuration.
7.	Simulation of Common Source JFET amplifier.
8.	Simulation of Tunnel Diode Characteristics.
Reference Books:	
1.	Lab manual
e-Resources:	
1.	http://vlabs.iitkgp.ac.in/be/#
2.	https://nptel.ac.in/courses/117/103/117103063/
3.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2106	PC	--	--	3	1.5	15	35	3 Hrs
DIGITAL LOGIC DESIGN LAB								
Course Objectives:								
1.	To outline the basics of Digital electronics, Boolean algebra, and be able to design the simple logic circuits and verify the functionality.							
2.	Design combinational and sequential logic circuits using digital ICs.							
3.	This laboratory course enables students to get practical experience in design, assembly and evaluation of switching theory and logic design. students use a digital trainer kit and Vivado emulator to test their electronic designs.							
Course Outcomes:On completion of this laboratory course, the students will be able to								
S.No	Outcome							KL
1.	Analyze and design basic combinational logic circuits using Digital IC’ and HDL programming.							K4
2.	Implement basic sequential logic circuits using Digital IC’s and HDL programming.							K3
LIST OF PROGRAMS								
A. HARDWARE								
1	Verify the basic logic gates and implement the following using basic logic gate ICs. a. Full adder. b. Full subtractor.							
2	Verify the logic functions of 8:1 Multiplexer using IC 74151 and 1:16 Demultiplexer using IC 74154. a. Implement the given function using 8:1 Multiplexer b. Convert 1:16 Demultiplexer into a 4:16 Decoder and implement the given function.							
3	Verify the given code convertors using digital trainer kit a. Implement BCD to Seven segment decoder driver using IC 74147 and Seven segment Display b. 8 to 3 Priority encoder using IC 74148.							
4	Verify the operation of following flip-flops using Digital Trainer Kit a. SR-latch & RS-latch b. JK- flip flop using IC 7476 along with direct inputs. c. D- flip flop d. T- flip flop							
5	Design and verify the following counters using Digital Trainer Kit a. Mod-16 Ripple counter using IC7493 b. Mod-8 Synchronous counter using 7476 and 7408 ICs c. Decade(Mod-10) counter using IC7493							
6	Verify the functioning of 5 bit shift register IC7496 using Digital Trainer Kit.							

B. SOFTWARE	
7	Verify the operation of following digital components using Vivado Simulator a. Full adder. b. Full subtractor.
8	Verify the operation of 8:1 multiplexer and 8:3 priority encoder using Vivado Simulator.
9	Design 4-bit ALU and verify some arithmetic and logical operations using Vivado Simulator.
10	Design Mod-16 ripple and Mod-8 synchronous counters using Vivado Simulator.
Reference Books:	
1.	Switching & Finite Automata theory – Zvi Kohavi and Neeraj K Jha, 3rd Edition, Cambridge.
2.	Digital Design – Morris Mano, PHI, 3rd Edition.
3.	Switching theory and logic design by Hill and Peterson, Mc-Graw Hill TMH edition, 2012.
4.	Digital ICs Lab Manual
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105185/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2107	PC	--	--	3	1.5	15	35	3hrs
SIGNALS AND SYSTEMS LAB								
(For ECE)								
Course Objectives:								
1.	Demonstrate the generation of various CT&DT signals and verify transformation operations using MATLAB.							
2.	Analyze the system characteristics using convolution and correlation operations, observe the behaviour of LTI systems with different input signals.							
3.	Illustrate the generation of spectrum for Continuous time periodic and non-periodic signals using Fourier series and Fourier transform respectively.							
4.	Demonstrate the effect of aliasing while generating and reconstruction of signals from continuous signals from discrete time signals and vice versa.							
Course Outcomes:On the completion of this laboratory course, the students will be able to								
1.	Construct various CT & DT signals and can perform basic operation on these signals using MAT LAB.							K4
2.	Analyse the spectral characteristics of continuous-time and discrete- time, periodic and aperiodic signals using Fourier analysis.							K4
3.	Analyse system properties based on impulse response and Fourier analysis							K4
4.	Classify systems based on their properties and determine the response of LSI system using convolution							K2
5.	Examine the process of sampling and the effects of under sampling.							K4
LIST OF PROGRAMS								
1	MATLAB Software & Toolboxes Introduction to MATLAB Basics, Commands, Matrix Algebra, Basic Vector Manipulation, a) Scripts, Functions, Graphics, b) Useful SPTB Functions							
2	Generation and Plotting of Basic Continuous-Time and Discrete-Time Signals. a) CT Signals: Sinusoidal, Exponential, Complex exponential, Unit impulse & Unit step functions b) DT Signals: Sinusoidal sequence, Exponential sequence, Complex exponential sequence, Unit impulse sequence & Unit step sequence c) Random Signals & Gaussian Noise							
3	Signal Transformations & Operations on Signals a) Amplitude scaling, DC biasing, b) Time shifting, Time reversal, Time scaling, c) Even/Odd decomposition d) Addition & Multiplication							

4	Convolution & Correlation of Signals a) CT Convolution b) DT Convolution c) Correlation d) Applications of Convolution & Correlation
5	Frequency Domain Representation of CT Periodic Signals: Fourier Series a) Fourier Series for CT Periodic Signals (FS): Computation of Fourier Coefficients & Plotting of Spectrum.
6	Frequency Domain Representation of CT Aperiodic Signals: Fourier Transform a) Fourier Transform for CT Aperiodic Signals (CTFT): CTFT Computation & Spectrum Plotting.
7	Frequency Domain Representation of DT Aperiodic Signals: DTFT a) Fourier Transform for DT Aperiodic Signals (DTFT): DTFT Computation & Spectrum Plotting
8	Frequency Domain Representation of LTI Systems a) Frequency Response of CT-LTI Systems b) Frequency Response of DT-LTI Systems
9	LTI System Simulation a) Response of a LTI system to different inputs: Sinusoidal, Unit impulse & Unit step signals
10	Sampling Theorem, Signal Reconstruction & Aliasing a) Sampling Theorem Verification & Observing Aliasing Effect.
11	Laplace Transform, Z-Transform & Inverse Transforms using MATLAB Symbolic Toolbox.
Reference Books:	
1.	Lab Manual

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2108	SOC	1	--	2	2	--	50	3 Hrs
ARDIUNO PROGRAMMING								
(For ECE)								
Course Objectives:								
1.	To provide knowledge of different smart applications.							
2.	To familiarize students with Arduino as IDE, programming language & platform.							
3.	To provide knowledge of Arduino boards and basic components.							
4.	Develop skills to design and implement various smart applications.							
Course Outcomes:On the completion of this laboratory course, the students will be able to								
S.No	Outcome							KL
1.	Learn the basics of electronics, including reading schematics (electronics diagrams)							K3
2.	Learn how to interface sensors with Arduino.							K3
3.	Analyze the Arduino Examples along with programming language and IDE.							K4
4.	Design Prototype circuits related to real world applications.							K4
LIST OF PROGRAMS								
1	Introduction to TINKERCAD Platform							
2	Introduction to Sensors & Other components.							
3	Introduction to Arduino Board and its Interfacing.							
4	Design a circuit to perform ON/OFF of LED with the help of Bread board.							
5	Design a circuit to perform Volume adjustment of Buzzer by using Potentiometer with the help of Bread board.							
6	Design a circuit to perform ON/OFF of LED by using Switch with the help of Bread board.							
7	Write a program to perform ON/OFF of LED and implement the same with the help of Arduino.							
8	Write a program to perform ON/OFF of LED by using Switch & implement the same with the help of Arduino.							
9	Write a program to print given text on LCD Display by interfacing it to Arduino.							
10	Write a program to interface LDR sensor with Arduino.							
11	Write a program to interface Ultrasonic sensor with Arduino.							
12	Assignments							
Reference Books:								
1.	http://mct.asu.edu.eg/uploads/1/4/0/8/14081679/lab1.pdf							
2.	Laboratory Manual for Embedded Controllers Using C and Arduino by James M. Fiore							
3.	http://wireless.ictp.it/Benin/Slides/Lab_1.pdf							
4.								
e-Resources:								
1.	https://www.tinkercad.com/							
2.	https://www.arduino.cc/							

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2101	MC	2	--	--	--	--	--	--
ENVIRONMENTAL SCIENCE								
(Common to CSE, ECE & IT)								
Course Objectives: The objectives of the course are to impart:								
1.	Overall understanding of the natural resources.							
2.	Basic understanding of the ecosystem and its diversity.							
3.	Acquaintance on various environmental challenges induced due to unplanned anthropogenic activities.							
4.	An understanding of the environmental impact of developmental activities.							
5.	Awareness on the social issues, environmental legislation and global treaties.							
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary nature of Environmental Studies: Definition, Scope and Importance – Sustainability: Stockholm and Rio Summit–Global Environmental Challenges: Global warming and climate change, acid rains, ozone layer depletion, population growth and explosion, effects;. Role of information technology in environment and human health. Ecosystems: Concept of an ecosystem. - Structure and function of an ecosystem; Producers, consumers and decomposers. - Energy flow in the ecosystem - Ecological succession. - Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems.							
UNIT-II (8 Hrs)	Natural Resources: Natural resources and associated problems. Forest resources: Use and over – exploitation, deforestation – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources: Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources. Food resources: World food problems, changes caused by non-agriculture activities-effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources. Land resources: Land as a resource, land degradation, Wasteland reclamation, man induced landslides, soil erosion and desertification; Role of an individual in conservation of natural resources; Equitable use of resources for sustainable lifestyles.							
UNIT-III (8 Hrs)	Biodiversity and its conservation: Definition: genetic, species and ecosystem diversity-classification - Value of biodiversity: consumptive use, productive use, social-Biodiversity at national and local levels. India as a mega-diversity nation - Hot-spots of biodiversity - Threats to biodiversity: habitat loss, man-wildlife conflicts. - Endangered and endemic species of India – Conservation of biodiversity: conservation of biodiversity.							

UNIT-IV (8 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Noise pollution, Nuclear hazards. Role of an individual in prevention of pollution. - Pollution case studies, Sustainable Life Studies. Impact of Fire Crackers on Men and his well being. Solid Waste Management: Sources, Classification, effects and control measures of urban and industrial solid wastes. Consumerism and waste products, Biomedical, Hazardous and e – waste management.
UNIT-V (8 Hrs)	Social Issues and the Environment: Urban problems related to energy -Water conservation, rain water harvesting-Resettlement and rehabilitation of people; its problems and concerns. Environmental ethics: Issues and possible solutions. Environmental Protection Act -Air (Prevention and Control of Pollution) Act. –Water (Prevention and control of Pollution) Act -Wildlife Protection Act -Forest Conservation Act-Issues involved in enforcement of environmental legislation.-Public awareness.
UNIT-VI (8 Hrs)	Environmental Management: Impact Assessment and its significance various stages of EIA, preparation of EMP and EIS, Environmental audit. Ecotourism, Green Campus – Green business and Greenpolitics. The student should Visit an Industry / Ecosystem and submit a report individually on any issues related to Environmental Studies course and make a power point presentation.
Text Books:	
1.	Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada Rani; Pearson Education, Chennai
2.	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.
3.	Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula
Reference Books:	
1.	Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2.	A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3.	Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4.	Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2103	MC	2	--	--	--	--	--	--
ENGLISH PROFICIENCY								
(Common to CSE, ECE & IT)								
Course Objectives: The students will be able to								
1.	Communicate their ideas and views effectively							
2.	Practice language skills and improve their language competency.							
3.	Know and perform well in real life contexts							
4.	Identify and examine their self-attributes which require improvement and motivation.							
5.	Build confidence and overcome their inhibitions, stage freight, nervousness etc.,							
6.	Improve their reading skills.							
Course Outcomes: On the completion of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Improve speaking skills.							K3
2.	Enhance their listening capabilities							K3
3.	Learn and practice the skills of composition writing.							K3
4.	Enhance their reading and understanding of different texts.							K3
5.	Improve their communication both in formal and informal contexts.							K3
6.	Be confident in presentation skills.							K3
SYLLABUS								
UNIT-I	Listening Skills Types of listening Hearing and Listening Listening as a receptive skill							
UNIT-II	Speaking Skills Presentation skills Describing event/place/thing Extempore Debate Group Discussion							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive) E-Mail Writing Business Letters Resume Writing							

UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking
Reference Books:	
1.	Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
2.	English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
3.	Enriching Speaking and Writing Skills, Orient Blackswan Publishers.
4.	The Oxford Guide to Writing and Speaking by John Seely OUP.
5.	Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.
6.	Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
Note: Internal Assessment is carried out throughout the semester.	

Regulation: R20		II / IV - B.Tech. II - Semester							
ELECTRONICS & COMMUNICATION ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20 EC 2201	Electronic Circuit Analysis and Design	PC	3	3	0	0	30	70	100
B20 EC 2202	Linear and Digital IC Applications	PC	3	3	0	0	30	70	100
B20 EC 2203	Analog Communications	PC	3	3	0	0	30	70	100
B20 EC 2204	Computer Architecture and Organization	PC	3	3	0	0	30	70	100
B20 EC 2205	Electromagnetic Field theory and Transmission Lines	PC	3	3	0	0	30	70	100
B20 EC 2206	Electronic Circuit Analysis And Design Lab With Simulation	PC	1.5	0	0	3	15	35	50
B20 EC 2207	Linear And Digital IC Applications Lab With Simulation	PC	1.5	0	0	3	15	35	50
B20 EC 2208	Analog Communication Lab with Simulation	PC	1.5	0	0	3	15	35	50
B20 EC 2209	Python Programming	SOC	2	1	0	2	--	50	50
TOTAL			21.5	20	0	11	195	505	700

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2201	PC	3	--	--	3	30	70	3 Hrs.
ELECTRONIC CIRCUIT ANALYSIS AND DESIGN								
(Common to ECE & EEE)								
Course Objectives:								
1.	Apply the concept of multistage amplifiers and analyze them.							
2.	Illustrate the analysis and design of feedback amplifiers, power amplifiers and tuned amplifiers.							
3.	Illustrate the principle of oscillator and analyze different types of sinusoidal oscillators							
4.	Illustrate the concept and analyze applications of op-amp.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
CO1	Outline the concepts of multistage amplifiers, feedback amplifiers, power amplifiers, tuned amplifiers, operational amplifiers and oscillators.							K2
CO2	Apply the concepts in the realization of practical circuits.							K3
CO3	Analyze and design practical electronic circuits using amplifiers, oscillators and operational amplifiers.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Multistage Amplifiers Transistor at high frequencies, CE short circuit current gain and concept of Gain Bandwidth product. BJT and FET RC coupled amplifiers at low and high frequencies. Frequency response and calculation of Bandwidth of Multistage Amplifiers.							
UNIT-II (10 Hrs)	Feedback Amplifiers Concept of Feedback Amplifiers - Effect of Negative Feedback on the amplifier Characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.							
UNIT-III (8 Hrs)	Sinusoidal Oscillators Condition for oscillations and types of Oscillators – RC Oscillators: RC Phase Shift and Wien bridge Oscillators. LC Oscillators: Hartley, Colpitts, Clapp, Tuned Collector and Crystal Oscillators.							
UNIT-IV (10 Hrs)	Power Amplifiers: Classification of Power Amplifiers. Series fed, Transformer coupled class-A and class-B Power amplifiers. Push Pull Class-A, Class-B and Class-AB Power Amplifiers. Cross-over Distortion in Pure Class-B Power Amplifier and Class-AB Power Amplifier-Trickle Bias, Derating Factor and Heat Sinks – Complementary Push Pull Amplifier. Tuned Voltage Amplifiers: Analysis of Single tuned, Double tuned and Stagger Tuned Amplifiers with gain and Bandwidth Calculations.							
UNIT-V (10 Hrs)	Operational Amplifiers: Concept of Differential Amplifier. Differential Amplifier supplied with a constant current source. Calculation of common mode rejection ratio. Block diagram							

	and Ideal characteristics of an Op-Amp. Applications of Op-Amp: Inverting and Non-Inverting amplifiers, Integrator, Differentiator, Summing, Subtracting and Logarithmic Amplifiers. Definition and Measurement of OP-Amp Parameters.
Text Books:	
1.	Integrated electronics-Jacob Millman and Christos C. Halkias
2.	Electronic Devices and Circuits – Robert L Boylestad & Lewis Nashalsky, 2009, 10th edition, Pearson publications.
3.	Op-amps and Linear Integrated Circuits – Gayakwad
Reference Books:	
1.	Electronic Devices and Circuits by Salivahanan. Tata McGraw-Hill Publication.
2.	Electronic Devices and Circuits by Sanjeev Gupta, 3rd Edition, Dhanpat Rai Publications
e-Resources:	
1.	https://www.electronics-tutorials.ws/oscillator/oscillators.html
2.	https://www.electronics-tutorials.ws/systems/negative-feedback.html
3.	https://www.uotechnology.edu.iq/dep-eee/lectures/3rd/Electronic/Analog%2520electronic/part2.pdf
4.	https://www.electronics-tutorials.ws/opamp/opamp_1.html

Code	Category	L	T	P	C	I.M	E.M	Exam	
B20EC2202	PC	3	--	--	3	30	70	3 Hrs	
LINEAR AND DIGITAL IC APPLICATIONS									
(For ECE)									
Course Objectives:									
1.	To design linear and non-linear applications of operational amplifier								
2.	To analyze the concepts of Active filters and waveform generators								
3.	To design the basic applications using 555 IC Timer								
4.	To analyze the concepts of Analog to digital and Digital to Analog converters								
5.	To analyze the concepts of various Digital Logic Families								
Course Outcomes:At the end of this course, the students will be able to									
S.No	Outcome								KL
1.	Design/analyze fundamental circuits based on op-amps								K4
2.	Design and analyze of various active filters, oscillators								K4
3.	Design and Analyze of various applications using IC 555 timer								K4
4.	Analyze the designing of Analog to digital and Digital to Analog converters								K4
5.	Analyze the various Digital IC Logic Families and to implement the logic function								K4
SYLLABUS									
UNIT-I (12 Hrs)	applications of operational amplifiers: Basics of Op-Amp, Block Diagram, Ideal characteristics, op-amp parameters, open loop, and closed loop op-amp configurations, Applications of op-amp- summing amplifier, subtractor, Basic Integrator, Basic differentiator, Logarithmic amplifier; Instrumentation Amplifiers, Op-amp As a Comparators, Schmitt trigger, square wave generator								
UNIT-II (9 Hrs)	active filters and oscillators: Butterworth type LPF, HPF first and secondorder filters, Higher order filters, Wide Band pass filters, Wide band reject filter, All pass filters. Op-Amp Phase Shift oscillator, Wein-bridge oscillator, Quadrature Oscillator, voltage-controlled oscillators, Voltage to current and current to voltage converters.								
UNIT-III (10 Hrs)	Special Ics Introduction, Block diagram, 555 timer as an Astable and Monostable Multivibrator, Applications of 555 Timer as Monostable multivibrator- Frequency divider, Pulse stretcher, Missing pulse detector, Linear ramp generator, Pulse width modulation, Applications of 555 Timer as Astable Multivibrator- Square wave oscillator, FSK Generator, Schmitt trigger.IC 565: Block diagram of PLL and applications of PLL IC 566 : Functional diagram of voltage controlled oscillator (IC 565)								
UNIT-IV (08 Hrs)	D to A and A to D CONVERTERS: Introduction, basic DAC techniques – weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, A to D converters - parallel comparator type ADC, counter type ADC, successive approximation ADC and dual slope ADC, DAC and ADC Specifications.								

UNIT-V (10 Hrs)	Digital logic Families: Introduction, TTL, ECL, MOS logic, CMOS logic-analysis CMOS LOGIC: CMOS logic levels, MOS transistors, Basic CMOS Inverter, NAND and NOR gates, CMOS AND-OR-INVERT and OR-AND-INVERT gates, implementation of logic functions using PAL and PLA
Text Books:	
1.	D. Roy Choudhury, Shail B. Jain (2012), Linear Integrated Circuit, 4th edition, New Age International Pvt.Ltd.,NewDelhi,India.
2.	Floyd, Jain (2009), Digital Fundamentals, 8th edition, Pearson Education, New Delhi.
Reference Books:	
1.	Ramakant A. Gayakwad, (2012), OP-AMP and Linear Integrated Circuits, 4th edition, Prentice Hall / Pearson Education, New Delhi.
e-Resources:	
1.	Op-Amp Practical Applications: Design, Simulation and Implementation https://nptel.ac.in/courses/108/108/108108114/
2.	Introduction to Combinational Circuits: https://www.youtube.com/watch?v=yt1Fwrc8ysk

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2203	PC	3	--	--	3	30	70	3 Hrs
ANALOG COMMUNICATIONS								
(For ECE)								
Course Objectives:								
1.	Familiarize with the fundamental concepts of communication systems and various techniques of Analog modulation and demodulation of signals.							
2.	To provide a good understanding of the behaviour of analog communications in the presence of noise							
3.	To classify and discuss different types of transmitters and receivers as applicable to analog communication systems							
4.	Familiarize with basic techniques of generating and demodulating Pulse modulated signals.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Analyze the concepts of Amplitude modulation and demodulation techniques.							K4
2.	Analyze the concepts of DSB-SC,SSB,VSB modulation and demodulation techniques.							K4
3.	Analyze the concepts of angle modulation and demodulation techniques.							K4
4.	Identify the functional blocks of transmitters and receivers							K2
5.	Compare the performance of analog modulation techniques in the presence of noise and differentiate various Pulse modulation and demodulation techniques.							K2
SYLLABUS								
UNIT-I (8 Hrs)	AMPLITUDE MODULATION: Introduction to communication system, Need for Frequency Translation, A Method for Frequency Translation, Amplitude Modulation, Definition, Time domain and frequency domain description, Single tone modulation, AM Modulators - Square law Modulator, Switching modulator, AM demodulation - Envelope detector, Square law demodulator, Maximum allowable modulation for rectifier detection, Spectrum and Power efficiency.							
UNIT-II (8 Hrs)	DSB-SC & SSB MODULATION: Double Side Band Suppressed Carrier (DSB-SC) Modulation, Time domain and frequency domain description, DSB-SC Modulators-Balanced Modulators, Ring Modulator, DSB-SC Demodulator-Coherent detection, Synchronous Detector, COSTAS Loop, SSB Modulation, SSB Modulators- Filter Method, Phase Shift Method, SSB Demodulator, Vestigial Side Band Modulation, Quadrature Amplitude Modulation, Frequency Division Multiplexing, Comparison of AM Techniques, Applications of different AM Systems.							
UNIT-III (8 Hrs)	ANGLE MODULATION: Angle Modulation, Phase and Frequency Modulation, Relationship between Phase and							

	Frequency Modulation, Phase and Frequency Deviation, Spectrum of an FM Signal, Bandwidth of Sinusoidally Modulated FM Signal, Effect of the Modulation Index β on Bandwidth, Spectrum of Constant Bandwidth FM, Phasor Diagram for FM Signals. FM Generation: Parameter variation method, Armstrong's Indirect method, Frequency Multiplication and application to FM, FM Demodulator, FM Demodulation using PLL, Zero crossing detector, Pre – emphasis and De – emphasis, Comparison of FM & AM.
UNIT-IV (8 Hrs)	TRANSMITTERS & RECEIVERS: Radio Transmitter - Classification of Transmitter, AM Transmitter, Effect of feedback on performance of AM Transmitter, FM Transmitter – Variable reactance type and phase modulated FM Transmitter, frequency stability in FM Transmitter. Radio Receiver - Receiver Types - Tuned radio frequency receiver, Super heterodyne receiver, RF section and Characteristics - Frequency changing and tracking, Intermediate frequency, AGC, FM Receiver, Comparison with AM Receiver, Amplitude limiting. Communication Receivers, extensions of super heterodyne principle and additional circuits.
UNIT-V (8 Hrs)	NOISE: Review of noise and noise sources, noise figure, Noise in Analog communication Systems Noise in DSB & SSB System, Noise in AM System, Noise in Angle Modulation Systems, Threshold effect in Angle Modulation System, Pre-emphasis & de-emphasis. PULSE MODULATION: Types of Pulse modulation, PAM (Single polarity, double polarity) PWM: Generation & demodulation of PWM, PPM, Generation and demodulation of PPM, Time Division Multiplexing, TDM Vs FDM
Text Books:	
1.	Principles of Communication Systems – H Taub & D. Schilling, Gautam Sahe, TMH, 3rd Edition, 2007.
2.	Principles of Communication Systems - Simon Haykin, John Wiley, 2nd Edition, 2007.
Reference Books:	
1.	Electronics & Communication System – George Kennedy and Bernard Davis, TMH 2004.
2.	Modern Digital and Analog Communication Systems –B.P.Lathi, Zhi Ding, Hari Mohan Gupta, Oxford University Press, 4th Edition, 2017
e-Resources:	
1.	https://nptel.ac.in/courses/117/105/117105143/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2204	PC	3	--	--	3	30	70	3Hrs
COMPUTER ARCHITECTURE AND ORGANIZATION								
(For ECE)								
Course Objectives:								
1.	Outline Basic building blocks of a Digital Computer and implementation of Computer architecture.							
2.	Apply the knowledge of microprogramming while designing CPU.							
3.	Analyze the concept of interfacing of peripheral devices & pipelining.							
4.	Outline the concept of Memories and design of Main memory.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze how computers represent and manipulates data							K3
2.	Develop the general architecture design of a digital computer							K3
3.	Acquiring the knowledge of designing microprograms for few basic instructions							K4
4.	Develop independent learning skills to interface main memory & I/O							K3
SYLLABUS								
UNIT-I (10 Hrs)	Register Transfer and Micro operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.							
UNIT-II (10 Hrs)	Basic Computer Organization: Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference Instructions, Input - Output and Interrupt, Complete Computer Description.							
UNIT-III (12 Hrs)	Micro programmed Control & CPU Organization: Micro programmed Control: Control Memory, Address Sequencing, Microinstruction Formats, Microprogram Example and Design of Control Unit. CPU Organization: Introduction, General Register Organization, Stack Organization Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer (RISC).							
UNIT-IV (12 Hrs)	Input – Output Organization & Introduction to Pipelining: Peripheral Devices, Input - Output Interface, Asynchronous Data Transfer, Modes of Transfer, Priority Interrupt, Direct Memory Access (DMA), Input- Output processor, CPU-IOP communication, Parallel Processing, Introduction to pipelining.							
UNIT-V (10 Hrs)	Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.							

Text Books:	
1.	Computer System Architecture, M. Morris Mano, PHI Publications, (3 rd Edition May 1996).
2.	Computer Architecture and Organization. John P.Hayes. McGraw Hill International, (3 rd Edition).
Reference Books:	
1.	Computer Organization, V. Carl Hamacher, Zvonko G. Vranesic and Safwat G. Zaky, McGraw Hill International, (4 th Edition).
2.	Digital Computer Fundamentals, Thomas C. Bartee.
e-Resources:	
1.	https://nptel.ac.in/content/storage2/courses/106103068/pdf/coa.pdf
2.	https://nptel.ac.in/courses/106/103/106103068/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2205	PC	3	--	--	3	30	70	3 Hrs
ELECTROMAGNETIC FIELD THEORY AND TRANSMISSION LINES								
(For ECE)								
Course Objectives:								
1.	To introduce the concepts of static electric field, steady magnetic field and time varying electromagnetic fields in real time applications.							
2.	To introduce Maxwell's equations and their applications in practical situations.							
3.	To introduce the fundamental theory of electromagnetic wave propagation in bounded and unbounded media.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate the behaviour of static electric and magnetic fields in different media for different charge and current distributions.							K3
2.	Apply Maxwell's equations to describe the behaviour of time varying electromagnetic fields.							K3
3.	Apply Maxwell's equations to describe the EM wave propagation in a homogeneous medium.							K3
4.	Compute different transmission line parameters.							K3
SYLLABUS								
UNIT-I (18 Hrs)		Electrostatics: Introduction, Coulomb's law and electric field intensity, electric field due to different types of charge distributions, Field due to infinite line charge and finite line charge, Field due to infinite sheet charge, electric flux density, gauss's law and applications, Energy and potential, electric field in terms of potential gradient, electric dipole, stored energy in static electric field and energy density, convection and conduction currents, continuity equation, conductors in electric field, relaxation time, dielectrics in electric field, Laplace's and Poisson's equations, Boundary conditions on <i>E</i> & <i>D</i> at the interface between two media.						
UNIT-II (12 Hrs)		Magnetostatics: Introduction, Biot-savart's law, Ampere's circuital law, applications of Ampere's circuital law, Point form of Ampere's circuital law, magnetic flux density, Gauss's law for magnetic fields, scalar and vector magnetic potentials, forces due to magnetic fields, magnetization in materials, boundary conditions on <i>H</i> & <i>B</i> at the interface between two media.						
UNIT-III (6 Hrs)		Maxwell's equations: Introduction, Faraday's law of electromagnetic induction, Maxwell's equations in integral and differential forms, Maxwell's equations using phasor notation, Boundary conditions on <i>E</i> , <i>D</i> , <i>H</i> & <i>B</i> at the interface between two media, Retarded Potentials.						

UNIT-IV (10 Hrs)	Electromagnetic Waves: Introduction, Wave equations for free space and for a conductive medium, uniform plane waves, properties of uniform plane waves, Relation between E and H in uniform plane wave, wave propagation in lossless and lossy media, Propagation in good conductors and good dielectrics, depth of penetration, polarization, Poynting's theorem.
UNIT-V (10 Hrs)	Transmission lines: Transmission lines - Introduction, types of transmission lines, equivalent circuit of transmission line, Primary and secondary constants of the line, Transmission line equations, characteristic impedance and expression for characteristic impedance, Reflection coefficient, standing wave ratio, lossless line, distortionless line, input impedance of transmission line, impedance matching with $\lambda/4$ line, reactance properties of shorted and open circuited lines of different lengths.
Text Books:	
1.	Principles of Electromagnetics - N.O.Sadiku, Oxford University Press, 4th edition.
2.	EM Waves and Radiating Systems – E.C.Jordan and K.G.Balmain, Printice Hall India
3.	Engineering Electromagnetics – W.A.Hayt and JABuck, Tata McGraw Hill.
Reference Books:	
1.	Electromagnetic Field Theory and Transmission Lines - GottapuSasibhushana Rao – Wiley
e-Resources:	
1.	https://nptel.ac.in/courses/108/104/108104087/
2.	https://www.youtube.com/watch?v=xrxO1FhUaWg&list=PLs5_Rtf2P2r5AJvc-EuBEwdur_dUVojW-

Code	Category	L	T	P	C	I.M	E.M	Exam	
B20EC2206	PC	--	--	3	1.5	15	35	3 Hrs.	
ELECTRONIC CIRCUIT ANALYSIS AND DESIGN LAB WITH SIMULATION									
(Common to ECE & EEE)									
Course Objectives:									
1.	To prepare students to perform the analysis of any Analog electronics circuit.								
2.	To empower students to understand the design and working of BJT / FET amplifiers, oscillators and Operational Amplifiers.								
3.	To evaluate the use of computer based analysis tools to review performance of semiconductor device circuits.								
4.	Model the electronic circuits using tools such as PSPICE and Multisim.								
Course Outcomes: On completion of this laboratory course, the students will be able to									
S.NO	Outcome								KL
1.	Apply the concepts of amplifier analysis to verify their characteristics and measure the important parameters.								K3
2.	Analyze the performance of power amplifiers.								K4
3.	Analyze the frequency response and characteristics of operational amplifiers.								K4
4.	Simulation and Design of different amplifiers and oscillator circuits.								K4
LIST OF EXPERIMENTS									
HARDWARE									
1	Frequency Response of RC Coupled Amplifier.								
2	Frequency Response of Negative Feedback Amplifier								
3	Colpitts Oscillator.								
4	RC Phase Shift Oscillator.								
5	Wein Bridge Oscillator.								
6	Hartley Oscillator.								
7	Basic Applications of Operational Amplifier.								
8	Tuned Voltage Amplifier.								
SOFTWARE									
1	Frequency Response of RC Coupled Amplifier.								
2	Frequency Response of Negative Feedback Amplifier								
3	Colpitts Oscillator.								
4	RC Phase Shift Oscillator.								
5	Wein Bridge Oscillator.								
6	Hartley Oscillator.								
7	Basic Applications of Operational Amplifier.								
8	Tuned Voltage Amplifier.								
Reference Books:									
1.	Lab manual.								
e-Resources:									
1.	https://www.orcad.com/resources/orcad-tutorials								

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2207	PC	--	--	3	1.5	15	35	3 Hrs.
LINEAR AND DIGITAL IC APPLICATIONS LAB WITH SIMULATION								
(For ECE)								
Course Objectives:								
1.	This laboratory course enables students to get practical experience in design, assembly, and evaluation of Linear and digital integrated circuits & its applications. They will use Multisim to test their electronic designs.							
Course Outcomes: On completion of this laboratory course, the students will be able to								
S.No	Outcome							KL
1.	Students will have a thorough understanding of operational amplifier (741)							K2
2.	Students will be able to design circuits using operational amplifiers for various applications							K4
3.	Know the differences between Linear and Digital Integrated IC's							K3
4.	Students will demonstrate their knowledge by designing analog circuits & digital circuits.							K4
List of Experiments								
Hardware								
1	Design of OP AMP Applications – (Implementation of Mathematical functions)							
2	Schmitt Trigger Circuits – using IC 741							
3	IC 741 Waveform Generators							
4	Active Filter Applications – LPF, HPF (first order)							
5	IC 555 Timer-Astable Multivibrator Circuits.							
6	Basic circuit of a flash type A/D converter							
Software								
1	Design of OP AMP Applications – (Implementation of Mathematical functions)							
2	Schmitt Trigger Circuits – using IC 741							
3	IC 741 Waveform Generators							
4	Active Filter Applications – LPF, HPF (first order)							
5	IC 555 Timer-Astable Multivibrator Circuits.							
6	Basic circuit of a flash type A/D converter							
e-Resources:								
1.	Op-Amp Practical Applications: Design, Simulation and Implementation https://nptel.ac.in/courses/108/108/108108114/							
2.	Electronic Modules for Industrial Applications using Op-Amps https://nptel.ac.in/courses/108/108/108108125/							
3.	Logic Gates:							

	Logic Gates https://www.youtube.com/watch?v=k9mMaHW-lV4&t=2s
4.	Full Adder https://www.youtube.com/watch?v=ryYh89GpPy0&t=614s Full adder
5.	Introduction to Combinational Circuits: https://www.youtube.com/watch?v=yt1Fwrc8ysk Introduction of Combinational Logic Circuits
6.	Full Subtractor https://www.youtube.com/watch?v=Gy18D6HSFaM

Code	Category	L	T	P	C	I.M	E.M	Exam	
B20EC2208	PC	--	--	3	1.5	15	35	3 Hrs	
ANALOG COMMUNICATION LAB WITH SIMULATION									
(For ECE)									
Course Objectives:									
1.	To implement various analog modulation and demodulation methods.								
2.	To be familiar with the design of various parameters of LPF, BPF and HPF.								
3.	To design IF and RF amplifiers and plot their frequency response.								
4.	To be familiar with different types of experiments like pre-emphasis, de-emphasis and DSB-SC waveform generators.								
Course Outcomes: On completion of this laboratory course, the students will be able to									
S.NO	Outcome								KL
1.	Design and implement modulation and demodulation circuits for amplitude modulation and frequency modulation techniques.								K4
2.	Design second order active filters for various frequency bands.								K4
3.	Construct the circuit and study the characteristics of different transmitter and receiver circuits such as Harmonic generator, RF Amplifier, pre-emphasis and de-emphasis								K4
LIST OF EXPERIMENTS									
1	Amplitude Modulation and Demodulation. (Hardware implementation, MATLAB simulation)								
2	Frequency Modulation and Demodulation. (MATLAB and Multisim simulation)								
3	Balanced Modulator. (Hardware implementation, MATLAB simulation)								
4	Harmonic Generator. (Hardware implementation)								
5	Pre-Emphasis and De-Emphasis. (Hardware implementation)								
6	RF Amplifier. (Hardware implementation)								
7	Design Of Active Low Pass and High Pass Filter. (Hardware implementation)								
8	Design Of Twin-T Network. (Hardware implementation)								
9	Pulse Modulation Techniques. (Hardware implementation, MATLAB simulation)								
10	Spectrum Analysis of AM & FM Signals. (MATLAB simulation)								

11	Frequency Division Multiplexing. (MATLAB simulation)
12	Simulation of AWGN Channel. (MATLAB simulation)
Reference Books:	
1.	Lab manual.
e-Resources:	
1.	https://nptel.ac.in/courses/117/105/117105143/
2.	https://nptel.ac.in/courses/117/101/117101106/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2209	SOC	1	--	2	2	--	50	3 hrs
PYTHON PROGRAMMING								
(For ECE)								
Course Objectives:								
1.	To introduce the basics of Python Programming Language.							
2.	To learn how to use lists, tuples, and dictionaries in Python programs							
3.	To learn how to write loops and decision statements in Python.							
Course Outcomes:On the completion of this laboratory course, the students will be able to								
S.NO	Outcome							KL
1.	Write, Test and Debug Python Programs							K3
2.	Implement Conditionals and Loops for Python Programs							K3
3.	Use functions and represent Compound data using Lists, Tuples and Dictionaries							K3
4.	Use python basic libraries in programming							K3
List of experiments								
1.	Basic programs: a. Python program to add two numbers b. Maximum of two numbers in Python c. Python program to print all Prime numbers in an Interval d. Python program to check whether a number is Prime or not e. Program to print ASCII Value of a character f. Python program to generate a random number g. Given a decimal number as input, the task is to write a Python program to convert the given decimal number into an equivalent binary number							
2.	Array Programs: a. Python Program to find sum of array b. Python Program to find largest element in an array c. Python Program for array rotation d. Python Program for Find reminder of array multiplication divided by n							
3.	List Programs: a. Python program to print even numbers in a list b. Python program to print all even numbers in a range c. Python program to find N largest elements from a list d. Python program to Sort the values of first list using second list							
4.	Matrix Programs a Adding and Subtracting Matrices in Python b. Python program to multiply two matrices							
5.	String Programs a. Python to find the frequencies of all the characters in that string and return a							

	dictionary with key as the character and its value as its frequency in the given string a. String slicing in Python to rotate a string b. Find words which are greater than given length k c. Python program to Check if a given string is binary string or not
6.	Dictionary Programs: a. Ways to sort list of dictionaries by values in Python – Using itemgetter b. Ways to sort list of dictionaries by values in Python – Using lambda function c. Python – Sort Dictionary key and values List
7	Tuple Programs: a. Python – Maximum and Minimum K elements in Tuple b. Create a list of tuples from given list having number and its cube in each tuple c. Python – Remove Tuples of Length K d. Sort a list of tuples by second Item
8	a. Python Program for Binary Search (Recursive and Iterative) b. Python Program for Linear Search
9.	NumPy, SciPy, matplotlib a. Basic mathematical operations- NumPy b. Compute pivoted LU decomposition of a matrix-SciPy c. Single Integrals-SciPy d. Matplotlib
Reference Books:	
1.	“Learning Python” by Mark Lutz and David Ascher 2nd Edition.
2.	“Think Python: How to Think Like a Computer Scientist”, by <u>Allen B. Downey</u> 2nd Edition .
e-Resources:	
1.	https://www.python.org/
2.	https://www.geeksforgeeks.org/python-programming-language/
3.	https://www.w3schools.com/python/default.asp